

STRUCTURAL ANALYSIS & DESIGN REPORT

Project:	Structural Analysis Engine Structural Model
Date:	2026-06-19 03:06:23
Analysis Type:	2D
Design Code:	AISC 360-16
Total Nodes:	8
Total Elements:	8

1. NODAL COORDINATES & REACTIONS

Node	X (m)	Y (m)	Z (m)	Rx (kN)	Ry (kN)	Rz (kN)	Mx	My	Mz
1	0.000	4.000	0.000	-0.00	-15.09	-	-	-	-6.79
2	4.000	4.000	0.000	-0.00	-107.31	-	-	-	0.00
3	10.000	4.000	0.000	0.00	-45.60	-	-	-	0.00
4	14.000	4.000	0.000	0.00	0.00	-	-	-	-0.00
5	2.240	8.300	0.000	0.00	0.00	-	-	-	0.00
6	6.130	8.930	0.000	0.00	0.00	-	-	-	0.00
7	10.220	8.320	0.000	0.00	0.00	-	-	-	0.00
8	13.250	6.460	0.000	0.00	0.00	-	-	-	0.00

2. ELEMENTS & SECTION PROPERTIES

Elem	Type	L (m)	A (m ²)	Iz (m ⁴)	Iy (m ⁴)	J (m ⁴)	fy (MPa)
1	FRAME	4.000	0.005	5e-05	5e-05	0.0001	250
2	FRAME	6.000	0.002124	1.34e-06	5e-05	0.0001	250
3	TRUSS	4.000	0.08343	0.004375	5e-05	0.0001	250
4	FRAME	4.848	0.005	5e-05	5e-05	0.0001	250
5	FRAME	3.941	0.005	5e-05	5e-05	0.0001	250
6	FRAME	4.135	0.005	5e-05	5e-05	0.0001	250
7	FRAME	3.555	0.005	5e-05	5e-05	0.0001	250
8	FRAME	2.572	0.005	5e-05	5e-05	0.0001	250

3. DESIGN CHECK RESULTS

Elem	Status	Ratio	Governing Criteria
1	FAIL	3.657	Bending (F) - Inelastic LTB
2	FAIL	71.073	Bending (F) - Compact (Lb <= Lp)
3	OK	0.000	Compression (E)
4	OK	0.000	Interaction (H1)
5	OK	0.000	Interaction (H1)
6	OK	0.000	Interaction (H1)
7	OK	0.000	Interaction (H1)
8	OK	0.000	Interaction (H1)

4. EQUILIBRIUM CHECK

[OK] Equilibrium Satisfied
 Sigma Fx = -7.8996e-10 kN
 Sigma Fy = +4.9158e-09 kN

Sigma Mz = +7.1981e-08 kNm

5. NODAL DISPLACEMENTS

Node	Ux (mm)	Uy (mm)	Uz (mm)	Rx (rad)	Ry (rad)	Rz (rad)
1	0.000	0.000	-	-	-	0.0000
2	0.000	0.000	-	-	-	0.0380
3	-0.000	-5.000	-	-	-	-0.3507
4	-0.000	0.000	-	-	-	0.0000
5	-0.000	0.000	-	-	-	-0.0000
6	-0.000	0.000	-	-	-	-0.0000
7	-0.000	-0.000	-	-	-	-0.0000
8	-0.000	-0.000	-	-	-	0.0000

6. MEMBER END FORCES (Local Axes)

Elem	Node	Nx	Vy	Mz
1	1	0.00	-15.09	-6.79
	2	-0.00	-64.91	106.41
2	2	0.00	-54.40	-76.41
	3	-0.00	-45.60	-0.00
3	3	0.00	0.00	0.00
	4	-0.00	0.00	0.00
4	5	-0.00	0.00	-0.00
	1	0.00	-0.00	0.00
5	6	-0.00	0.00	0.00
	5	0.00	-0.00	0.00
6	7	-0.00	0.00	0.00
	6	0.00	-0.00	-0.00
7	8	-0.00	-0.00	0.00
	7	0.00	0.00	-0.00
8	4	-0.00	-0.00	-0.00
	8	0.00	0.00	-0.00

7. GLOBAL MATRICES

GLOBAL FORCE VECTOR {F}:

[0, 40, 40, 0, 68.5, 42.1, 0, 59.5, -59, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

GLOBAL STIFFNESS MATRIX [K]:

0	0	0	0	0	0	0	0	-4.49e+04	-8.41e+04	-2.26e+03	0	0	0	0	0	0	0	0	0
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28e+02	6.56e+02	0	0	0	0	0	0	-8.41e+04	-1.62e+05	1.18e+03	0	0	0	0	
56e+02	8.75e+02	0	0	0	0	0	0	2.26e+03	-1.18e+03	4.13e+03	0	0	0	0	
0	0	-7.08e+04	0	0	0	0	0	0	0	0	0	0	0	0	
43e+02	-6.12e+02	0	-1.49e+01	4.47e+01	0	0	0	0	0	0	0	0	0	0	
12e+02	1.93e+03	0	-4.47e+01	8.93e+01	0	0	0	0	0	0	0	0	0	0	
0	0	4.24e+06	0	0	-4.17e+06	0	0	0	0	0	0	0	0	0	
49e+01	-4.47e+01	0	1.49e+01	-4.47e+01	0	0	0	0	0	0	0	0	0	0	
47e+01	8.93e+01	0	-4.47e+01	1.79e+02	0	0	0	0	0	0	0	0	0	0	
0	0	-4.17e+06	0	0	4.21e+06	-1.06e+05	-8.68e+03	0	0	0	0	0	0	0	
0	0	0	0	0	-1.06e+05	3.56e+05	-2.65e+03	0	0	0	0	0	0	0	
0	0	0	0	0	-8.68e+03	-2.65e+03	1.56e+04	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	2.92e+05	1.24e+05	1.65e+03	-2.47e+05	-3.97e+04	-6.18e+02	0	
0	0	0	0	0	0	0	0	1.24e+05	1.71e+05	2.63e+03	-3.97e+04	-8.40e+03	3.81e+03	0	
0	0	0	0	0	0	0	0	1.65e+03	2.63e+03	1.84e+04	6.18e+02	-3.81e+03	5.08e+03	0	
0	0	0	0	0	0	0	0	-2.47e+05	-3.97e+04	6.18e+02	4.84e+05	4.70e+03	1.14e+03	-2.37e+05	3.50e
0	0	0	0	0	0	0	0	-3.97e+04	-8.40e+03	-3.81e+03	4.70e+03	1.53e+04	-3.44e+02	3.50e+04	-6.92e
0	0	0	0	0	0	0	0	-6.18e+02	3.81e+03	5.08e+03	1.14e+03	-3.44e+02	1.98e+04	-5.18e+02	-3.47e
0	0	0	0	0	0	0	0	0	0	0	-2.37e+05	3.50e+04	-5.18e+02	4.42e+05	-1.59e
0	0	0	0	0	0	0	0	0	0	0	3.50e+04	-6.92e+03	-3.47e+03	-1.59e+05	8.58e
0	0	0	0	0	0	0	0	0	0	0	5.18e+02	3.47e+03	4.84e+03	1.97e+03	5.75e
0	0	0	0	0	-3.95e+04	1.06e+05	8.68e+03	0	0	0	0	0	0	-2.05e+05	1.24e
0	0	0	0	0	1.06e+05	-3.56e+05	2.65e+03	0	0	0	0	0	0	1.24e+05	-7.89e
0	0	0	0	0	-8.68e+03	-2.65e+03	7.78e+03	0	0	0	0	0	0	2.48e+03	4.05e

10. ANALYSIS WARNINGS

- Element 1 [AISC 360-16 (LRFD)]: Bending (F) - Inelastic LTB (Utilization = 3.66)
- Element 2 [AISC 360-16 (LRFD)]: Bending (F) - Compact (Lb <= Lp) (Utilization = 71.07)